

AYMANE AIT DADS

Generative AI Engineer Intern | LLM Fine-Tuning · RAG & Agentic Workflows · Prompt Engineering
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PROFESSIONAL SUMMARY

Achieved top 10% on the NVIDIA Nemotron Reasoning Kaggle Competition (0.80/1.0, \$106K+ prize pool) by fine-tuning a 30B Mamba-Transformer MoE with LoRA and conducting 20+ single-variable ablations - directly demonstrating LLM fine-tuning, prompt engineering, and experimental methodology required for SAP GenAI CoE. Ranked 1st in EURECOM class on an AI-generated image detection leaderboard (0.791 AUC) and submitted to the NTIRE 2026 @ CVPR CodaBench international benchmark, applying ensemble methods and transfer learning at scale on 250K training images. Brings proficiency in Python, Hugging Face Transformers, vLLM, RAG pipelines, agentic workflows, and REST APIs across real-world generative AI projects. Available full-time June/July 2026 through December 2026 for a 5-6 month internship in Singapore; no visa sponsorship constraint anticipated.

CORE COMPETENCIES

LLM Fine-Tuning | Prompt Engineering | Agentic Workflows | RAG Pipelines | vLLM Deployment | Python-based AI/ML | Generative AI | Ablation Studies

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Developed a clustering pipeline using Random Forest and K-Means on **100K+ fiber-optic network records** to classify quality across thousands of nodes and map customer profiles to 5 commercial service tiers.
- Reduced manual analysis time by **~60%** through automated feature engineering and model evaluation pipelines, freeing the network optimization team for higher-value maintenance tasks.
- Delivered stakeholder presentations translating **model outputs into actionable maintenance recommendations**, communicating complex ML results to non-technical business partners.
- Built and evaluated ensemble models with Scikit-learn and Pandas, producing **Power BI dashboards** consumed directly by the network optimization team for operational decisions.

PROJECTS

NVIDIA Nemotron Reasoning Challenge

Kaggle Competition · In Progress June 2026 · Top 10% Leaderboard · \$106K+ Prize Pool

- Achieved **top 10% public leaderboard score (0.80/1.0)** on a \$106K+ Kaggle competition by fine-tuning Nemotron-3-Nano-30B, a hybrid 30B Mamba-Transformer MoE, with a LoRA adapter ($r=32$, ~888M trainable parameters) using SFTTrainer on 7,828 chain-of-thought reasoning examples.
- Resolved **5 sequential Blackwell GPU (sm_120) incompatibilities** - including monkey-patched allocator warmup, Mamba CUDA kernel disabling, and a pure-PyTorch RMSNorm fallback - and conducted 20+ single-variable ablations confirming completion-only loss and packing strategy impact on OOD generalization.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Mamba · Triton

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR CodaBench · 2025 · Ranked 1st in Class

- Ranked **1st in the EURECOM class private Kaggle leaderboard (0.791 AUC)** by fine-tuning a CLIP ViT-L/14 backbone (428M params) with a custom MLP head, dual learning rates, and strategic unfreezing of up to 8 transformer blocks trained on 250K images across 25+ generator types.
- Submitted to **NTIRE 2026 @ CVPR CodaBench** international benchmark; applied a 3-model ensemble weighted by validation AUC and 10-view test-time augmentation, discovering that 1-epoch training (0.793 AUC) outperformed 4-epoch training (0.767 AUC) due to superior out-of-distribution generalization.

PyTorch · OpenCLIP · Hugging Face Transformers · FP16 Mixed Precision · Kaggle T4 GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Cloud Computing, Statistics, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

LLM Fine-Tuning & Generative AI: LoRA / QLoRA | SFTTrainer (Unsloth) | vLLM | Prompt Engineering | RAG Pipelines | Agentic Workflows | LangChain

Python-based AI/ML & Frameworks: PyTorch | Hugging Face Transformers | PEFT / TRL | Scikit-learn | OpenCLIP | Pandas / NumPy | Git

Infrastructure & APIs: REST APIs | Docker (basic) | Supabase / PostgreSQL | Linux / Bash | Power BI | CUDA Debugging | Triton